

Demonstration of JetUnit: Rendering Diverse Force Feedback in Virtual Reality Using Water Jets

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ABSTRACT

JetUnit [21] is a water-based VR haptic system capable of generating a diverse range of perceived force intensities and frequencies through water jets. A key challenge in designing this system was optimizing parameters to enable the haptic device to produce force feedback that closely mimics the most intense force achievable with direct water jets, while ensuring the user remains dry. In this demonstration, we showcase JetUnit by integrating our haptic solutions into various virtual reality interactions, including touch, poking, injection, and recurring pulsing and wave signals.

CCS CONCEPTS

• **Human-centered computing** → **Haptic devices**; **Human computer interaction (HCI)**; **Interaction devices**.

KEYWORDS

haptics, water jets, force feedback, VR

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1 INTRODUCTION

Force feedback plays an important role in enhancing the immersiveness of virtual reality (VR) environments. Studies have demonstrated that synchronizing realistic force feedback with visual rendering substantially improves user engagement and the overall VR experience [14]. Recent research has explored various mechanisms for delivering force feedback in VR, including pneumatic systems [1, 2, 5, 6, 8, 16], exoskeletons [3, 13, 15, 17–20], electric muscle stimulation (EMS) [4, 7, 9, 11], and combinations thereof [10, 12].

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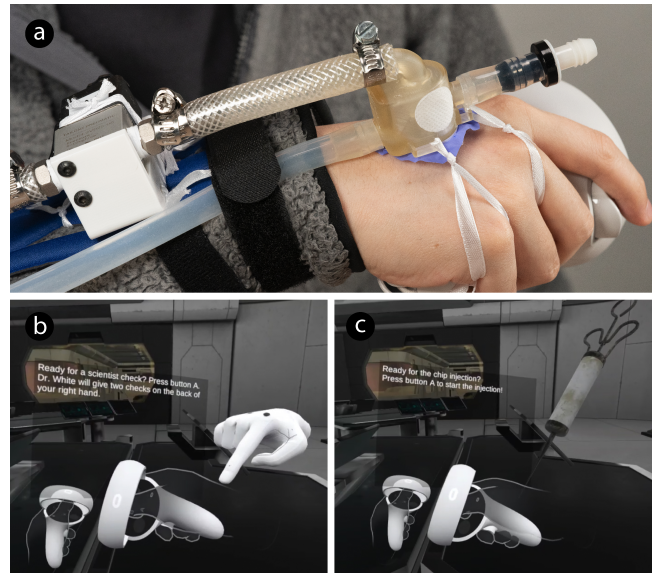


Figure 1: (a) The JetUnit offers force feedback over a wide range of perceived force intensities and pulsing frequencies. A single wearable JetUnit can render interactions ranging from (b) a gentle touch to (c) a progressively accelerating injection.

While each of these approaches can effectively deliver certain types of force feedback, they are often limited by the strength, frequency, or pattern of the force that can be produced, thus constraining the range of VR application scenarios.

To address this challenge, we propose *JetUnit* [21], a working wearable prototype engineered to provide force feedback across a broad spectrum of perceived force intensities and frequencies for various VR scenarios. Central to our system is the use of water jets at varying speeds to provide force feedback, enabling the delivery of both strong and gentle sensations in various patterns on the user's body, similar to those experienced with water-based massage systems like Jacuzzis. The key to JetUnit implementation is its custom-designed chamber unit. The chamber unit propels water directly onto a thin membrane, which transmits haptic sensations to the user's skin. The membrane, securely sealed at the chamber's

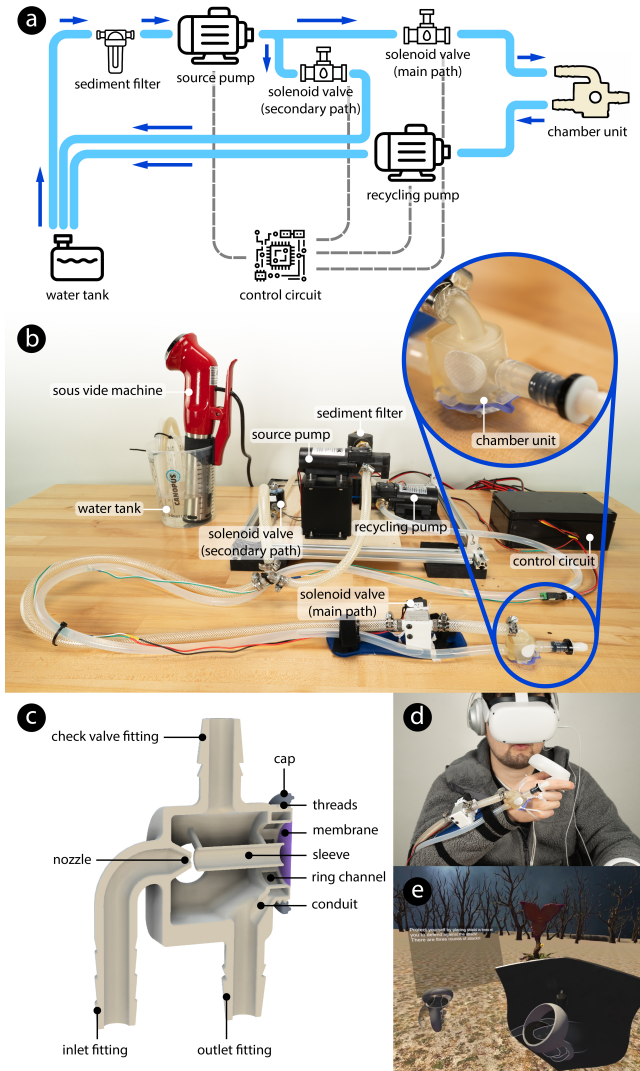


Figure 2: (a) Schematic of JetUnit System. (b) The JetUnit system setup. (c) Perspective view of the chamber unit's cross-section. A user is wearing JetUnit device as (d), defending attacks from a flower-shaped monster in VR (e), with pollens hitting the shield with different frequencies.

opening, ensures the water remains contained. The current JetUnit prototype can achieve a range of 16 kPa to 442 kPa on-skin contact pressure and a maximum frequency of 10 FPS. In this demonstration, we will let participants experience various haptic patterns in VR enabled by our single-unit water-based haptic solution.

2 IMPLEMENTATION

Figure 2a illustrates the JetUnit schematic. It includes four major components: a water source with a tank and water pumps, a network of tubing that transports water to and from the chamber for actuation and recycling, a chamber unit that provides force feedback, and a control system.

2.1 Chamber Unit Implementation

The custom 3D-printed chamber unit is designed to render force feedback. It serves two main functions: providing force feedback with varying intensities and frequencies through water jets, while also keeping the user dry.

As shown in Figure 2c, the chamber unit is a 25 mm × 28 mm × 32 mm quasi-cylindrical container with an opening at the front covered by a thin elastic membrane. The water stream flowing through the tubing network reaches the chamber. It then passes through a 1.2 mm diameter nozzle, hitting the membrane and delivering force feedback to the user.

While the membrane contains the water and keeps the user from getting wet, it causes water to accumulate within the chamber, leading to turbulence that reduces the intensity of the rendered force. To maximize the rendered force, we connected the chamber outlet to a recycling pump, ensuring efficient water drainage. Additionally, we incorporated three internal features into the chamber unit. First, a 3D-printed sleeve structure is positioned directly in front of the nozzle as Figure 2c. The pressurized water jets from the nozzle will thus first enter the sleeve before coming into contact with any residual water inside the chamber. Second, a ring channel with side openings is added directly to the membrane area. This design enables rapid evacuation of water from the chamber surrounding the membrane area to the outlet of the chamber. Finally, a check valve and two hydrophobic PTFE adhesive patches are installed on the chamber's sidewalls to balance the air pressure within the chamber. These features collectively ensure that no stranded water blocks the jetted water from the nozzle to the membrane.

3 VR DEMO

We will present a VR story centered on the experience with JetUnit. The story is structured around a single mission, divided into various scenes. Demo participants will be cast as 'the chosen' warrior, equipped with 'magic' technology by a group of scientists to defeat a flower-shaped monster. When exploring in VR, participants will wear the JetUnit on the dorsal side of their hands and feel various forces and patterns. For example, in one scene, participants will choose from three different shields to protect themselves from the flower-shaped monster. These shields vary in appearance. As participants explore the shield options, distinct haptic feedback patterns, such as sine waves, square waves, or sawtooth waves, are activated. In another scene, participants will receive a needle injection to implant 'magic' enhancement fluid beneath the skin on the back of their hands. Wearing the JetUnit, participants will feel an increasing force rendered to their hand during the 'injection'.

4 CONCLUSION

JetUnit is a haptic solution that utilizes a water system to create a diverse range of haptic patterns. This system achieves a broad spectrum of perceived force intensities and pulsing frequencies for haptic feedback. It also enables the rendering of force feedback with gradually changing magnitudes on the body, opening new possibilities for enhancing VR immersion. We will demonstrate the feasibility of JetUnit with a live demo at UIST 2024.

REFERENCES

- [1] Kyung Yun Choi, Neska Elhaoui, Jinmo Lee, Rosalind W. Picard, and Hiroshi Ishii. 2022. Design and Evaluation of a Clippable and Personalizable Pneumatic-haptic Feedback Device for Breathing Guidance. *Proc. ACM Interact. Mob. Wearable Ubiquitous Technol.* 6, 1, Article 6 (mar 2022), 36 pages. <https://doi.org/10.1145/3517234>
- [2] Alexandra Delazio, Ken Nakagaki, Roberta L. Klatzky, Scott E. Hudson, Jill Fain Lehman, and Alanson P. Sample. 2018. Force Jacket: Pneumatically-Actuated Jacket for Embodied Haptic Experiences. In *Proceedings of the 2018 CHI Conference on Human Factors in Computing Systems* (Montreal QC, Canada) (CHI '18). Association for Computing Machinery, New York, NY, USA, 1–12.
- [3] Atena Fadaei Jouybari, Kenny Jeanmonod, Olivier A. Kannape, Jevita Potheegadoo, Hannes Bleuler, Masayuki Hara, and Olaf Blanke. 2022. Cogno-Vest: A Torso-Worn, Force Display to Experimentally Induce Specific Hallucinations and Related Bodily Sensations. *IEEE Transactions on Cognitive and Developmental Systems* 14, 2 (2022), 497–506. <https://doi.org/10.1109/TCDS.2021.3051395>
- [4] Farzam Farbiz, Zhou Hao Yu, Corey Manders, and Waqas Ahmad. 2007. An electrical muscle stimulation haptic feedback for mixed reality tennis game. In *ACM SIGGRAPH 2007 Posters* (San Diego, California) (SIGGRAPH '07). Association for Computing Machinery, New York, NY, USA, 140–es. <https://doi.org/10.1145/1280720.1280873>
- [5] Sebastian Günther, Dominik Schön, Florian Müller, Max Mühlhäuser, and Martin Schmitz. 2020. PneuMolley: Pressure-based Haptic Feedback on the Head through Pneumatic Actuation. In *Extended Abstracts of the 2020 CHI Conference on Human Factors in Computing Systems* (Honolulu, HI, USA) (CHI EA '20). Association for Computing Machinery, New York, NY, USA, 1–10. <https://doi.org/10.1145/3334480.3382916>
- [6] Liang He, Cheng Xu, Ding Xu, and Ryan Brill. 2015. PneuHaptic: delivering haptic cues with a pneumatic armband. In *Proceedings of the 2015 ACM International Symposium on Wearable Computers* (Osaka, Japan) (ISWC '15). Association for Computing Machinery, New York, NY, USA, 47–48. <https://doi.org/10.1145/2802083.2802091>
- [7] Satoshi Hosono, Tamon Miyake, Shota Miyake, and Emi Tamaki. 2022. Feedback Method of Force Controlled by Electrical Muscle Stimulation Based on Infrared Optical Sensing. *Frontiers in Virtual Reality* 3 (2022), 880238. <https://doi.org/10.3389/frvir.2022.880238>
- [8] Smita Kanjanapas, Cara M. Nunez, Sophia R. Williams, Allison M. Okamura, and Ming Luo. 2019. Design and Analysis of Pneumatic 2-DoF Soft Haptic Devices for Shear Display. *IEEE Robotics and Automation Letters* 4, 2 (2019), 1365–1371. <https://doi.org/10.1109/LRA.2019.2895890>
- [9] Yuichi Kurita, Takaaki Ishikawa, and Toshio Tsuji. 2016. Stiffness Display by Muscle Contraction Via Electric Muscle Stimulation. *IEEE Robotics and Automation Letters* 1, 2 (2016), 1014–1019. <https://doi.org/10.1109/LRA.2016.2529689>
- [10] Pedro Lopes, Alexandra Ion, and Patrick Baudisch. 2015. Impacto: Simulating Physical Impact by Combining Tactile Stimulation with Electrical Muscle Stimulation. In *Proceedings of the 28th Annual ACM Symposium on User Interface Software & Technology* (Charlotte, NC, USA) (UIST '15). Association for Computing Machinery, New York, NY, USA, 11–19. <https://doi.org/10.1145/2807442.2807443>
- [11] Pedro Lopes, Sijing You, Lung-Pan Cheng, Sebastian Marwecki, and Patrick Baudisch. 2017. Providing Haptics to Walls & Heavy Objects in Virtual Reality by Means of Electrical Muscle Stimulation. In *Proceedings of the 2017 CHI Conference on Human Factors in Computing Systems* (Denver, Colorado, USA) (CHI '17). Association for Computing Machinery, New York, NY, USA, 1471–1482. <https://doi.org/10.1145/3025453.3025600>
- [12] Romain Nith, Shan-Yuan Teng, Pengyu Li, Yujie Tao, and Pedro Lopes. 2021. DextrEMS: Increasing Dexterity in Electrical Muscle Stimulation by Combining it with Brakes. In *The 34th Annual ACM Symposium on User Interface Software and Technology* (Virtual Event, USA) (UIST '21). Association for Computing Machinery, New York, NY, USA, 414–430. <https://doi.org/10.1145/3472749.3474759>
- [13] Evan Pezent, Priyanshu Agarwal, Jessica Hartcher-O'Brien, Nicholas Colonese, and Marcia K. O'Malley. 2022. Design, Control, and Psychophysics of Tasbi: A Force-Controlled Multimodal Haptic Bracelet. *IEEE Transactions on Robotics* 38, 5 (2022), 2962–2978. <https://doi.org/10.1109/TRO.2022.3164840>
- [14] Paul Richard, Georges Birebent, Philippe Coiffet, Grigore Burdea, Daniel Gomez, and Noshir Langrana. 1996. Effect of Frame Rate and Force Feedback on Virtual Object Manipulation. *Presence: Teleoperators and Virtual Environments* 5, 1 (02 1996), 95–108. <https://doi.org/10.1162/pres.1996.5.1.95>
- [15] Samuel B. Schorr and Allison M. Okamura. 2017. Fingertip Tactile Devices for Virtual Object Manipulation and Exploration. In *Proceedings of the 2017 CHI Conference on Human Factors in Computing Systems* (Denver, Colorado, USA) (CHI '17). Association for Computing Machinery, New York, NY, USA, 3115–3119. <https://doi.org/10.1145/3025453.3025744>
- [16] Aishwari Talhan, Yongjae Yoo, and Jeremy R. Cooperstock. 2024. Soft Pneumatic Haptic Wearable to Create the Illusion of Human Touch. *IEEE Transactions on Haptics* 17, 2 (2024), 177–190. <https://doi.org/10.1109/TOH.2023.3305495>
- [17] Shan-Yuan Teng, Pengyu Li, Romain Nith, Joshua Fonseca, and Pedro Lopes. 2021. Touch&Fold: A Foldable Haptic Actuator for Rendering Touch in Mixed Reality. In *Proceedings of the 2021 CHI Conference on Human Factors in Computing Systems* (Yokohama, Japan) (CHI '21). Association for Computing Machinery, New York, NY, USA, Article 736, 14 pages. <https://doi.org/10.1145/3411764.3445099>
- [18] Hsin-Ruey Tsai and Bing-Yu Chen. 2019. ElastImpact: 2.5D Multilevel Instant Impact Using Elasticity on Head-Mounted Displays. In *Proceedings of the 32nd Annual ACM Symposium on User Interface Software and Technology* (New Orleans, LA, USA) (UIST '19). Association for Computing Machinery, New York, NY, USA, 429–437. <https://doi.org/10.1145/3332165.3347931>
- [19] Hsin-Ruey Tsai, Yu-So Liao, and Chieh Tsai. 2022. ImpactVest: Rendering Spatio-Temporal Multilevel Impact Force Feedback on Body in VR. In *Proceedings of the 2022 CHI Conference on Human Factors in Computing Systems* (New Orleans, LA, USA) (CHI '22). Association for Computing Machinery, New York, NY, USA, Article 356, 11 pages. <https://doi.org/10.1145/3491102.3501971>
- [20] Hsin-Ruey Tsai, Jun Rekimoto, and Bing-Yu Chen. 2019. ElasticVR: Providing Multilevel Continuously-Changing Resistive Force and Instant Impact Using Elasticity for VR. In *Proceedings of the 2019 CHI Conference on Human Factors in Computing Systems* (Glasgow, Scotland UK) (CHI '19). Association for Computing Machinery, New York, NY, USA, 1–10. <https://doi.org/10.1145/3290605.3300450>
- [21] Zining Zhang, Jiasheng Li, Zeyu Yan, Jun Nishida, and Huaishu Peng. 2024. JetUnit: Rendering Diverse Force Feedback in Virtual Reality Using Water Jets. In *The 37th Annual ACM Symposium on User Interface Software and Technology* (Pittsburgh, PA, USA) (UIST '24). Association for Computing Machinery, New York, NY, USA, 15 pages. <https://doi.org/10.1145/3654777.3676440>